$$2y^4\frac{\partial z}{\partial x}-xy\frac{\partial z}{\partial y}=x\sqrt{z^2+1}.$$

$$\frac{2y^4}{x}\cdot\frac{\partial z}{\partial x}-y\frac{\partial z}{\partial y}=\sqrt{z^2+1}.$$

$$\frac{x\,dx}{2y^4}=\frac{dy}{-y}=\frac{dz}{\sqrt{z^2+1}}.$$

1) $x\,dx=-2y^3\,dy.\ x^2+y^4=A.$

2) $\dfrac{dy}{-y}=\dfrac{dz}{\sqrt{z^2+1}}.\ \operatorname{arsh}z+\ln|y|=B.$

В результате $F\left(x^2+y^4;\operatorname{arsh}z+\ln|y|\right)=0.\ \operatorname{arsh}z+\ln|y|=f\left(x^2+y^4\right).$

$z=\operatorname{sh}\left(f\left(x^2+y^4\right)-\ln|y|\right).$